

Расчетная графическая работа «Приложение определенного интеграла»

Задание 1 Найти площадь фигуры, ограниченной указанными линиями. Выполнить чертеж.

Вар-т	Задание	Вар-т	Задание
1	$a) y = x^2 + 3x - 7, y = -x - 7;$ $b) y = 4 - x^2, y = 0;$ $c) y = (x - 2)^2, y = 4x - 8$	2	$a) y = x^2 + 3x - 5, y = x + 3;$ $b) xy = 4, x = 1, x = 4, y = 0;$ $c) y = x\sqrt{9 - x^2}, y = 0, (0 \leq x \leq 3)$
3	$a) y = -x^2 + x, y = x - 9;$ $b) y^2 = 2x + 4, x = 0;$ $c) y = 4 - x^2, y = x^2 - x$	4	$a) y = -x^2 + 2x - 4, y = -3x;$ $b) y = x^2, y = 2 - x^2;$ $c) y = \sin x \cos^2 x, y = 0, \left(0 \leq x \leq \frac{\pi}{2}\right)$
5	$a) y = x^2 + 5x - 1, y = -2x^2 - x - 1;$ $b) y = \ln x, x = e, y = 0;$ $c) y = \sqrt{4 - x^2}, y = 0, x = 0, x = 1$	6	$a) y = x^2 - 2x - 2, y = -x;$ $b) y = x^3, x = 0, y = 8;$ $c) y = x^2\sqrt{4 - x^2}, y = 0, (0 \leq x \leq 2)$
7	$a) y = 2x^2 + 2x, y = -x^2 + 1;$ $b) y = \sin x, x \in [0, \pi], y = 0;$ $c) y = \sin^2 x \cos x,$ $y = 0, \left(0 \leq x \leq \frac{\pi}{2}\right)$	8	$a) y = x^2 + 5x, y = 4x + 6;$ $b) xy = 6, x + y = 7;$ $c) y = \sqrt{e^x - 1}, y = 0, x = \ln 2$
9	$a) y = -x^2 + 8x + 3, y = 5x + 3;$ $b) y = x, y = 2x, x = 3;$ $c) y = 1/(x\sqrt{1 + \ln x}),$ $y = 0, x = 1, x = e^3$	10	$a) y = -x^2 - 3x + 2, y = -6 - x;$ $b) y = x, y = 3x, x = 2;$ $c) y = \arccos x, y = 0, x = 0$
11	$a) y = x^2 + 4x - 3, y = -x^2 + 4x + 5;$ $b) y = 1 - x^2, y = 0;$ $c) y = (x + 1)^2, y^2 = x + 1$	12	$a) y = x^2 + 4x + 11, y = -x^2 - x + 9;$ $b) y = 3 - x^2, x = 0, y = 0;$ $c) y = 2x - x^2 + 3, y = x^2 - 4x + 3$
13	$a) y = x^2 + 7x + 2, y = x - 6;$ $b) y = x, x = 2, y = 0;$ $c) y = x\sqrt{36 - x^2},$ $y = 0, (0 \leq x \leq 6)$	14	$a) y = x^2 - 13x, y = -10x;$ $b) y = e^x, y = e^{2x}, x = 1;$ $c) x = \arccos y, y = 0, x = 0$
15	$a) y = x^2 + 2x - 3, y = 1 - x;$ $b) y = 2^x, y = 2^{2x}, x = 1;$ $c) y = x \arctg x, y = 0, x = \sqrt{3}$	16	$a) y = -x^2 + 6x - 6, y = -x + 4;$ $b) y = \cos x, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], y = 0;$ $c) y = x^2\sqrt{8 - x^2},$ $y = 0, (0 \leq x \leq 2\sqrt{2})$

17	$a) y = -x^2 - x + 17, y = x + 17;$ $b) y = x^2, y^2 = 4 - x^2;$ $c) x = \sqrt{e^y - 1}, x = 0, y = \ln 2$	18	$a) y = -x^2 + 2x, y = -3;$ $b) xy = 3, x = 1, x = 3, y = 0;$ $c) y = x\sqrt{4 - x^2},$ $y = 0, (0 \leq x \leq 2)$
19	$a) y = x^2 - 1, y = -x^2 + 17;$ $b) y^2 = x + 1, x = 0;$ $c) y = x/(1 + \sqrt{x}), y = 0, x = 1$	20	$a) y = x^2 + x - 12, y = -12;$ $b) y^2 = x, y^2 = 2x, x = 1;$ $c) y = \frac{1}{1 + \cos x}, y = 0, x = \frac{\pi}{2}, x = -\frac{\pi}{2}$
21	$a) y = x^2 - 4x + 8, y = -2x^2 - 3x + 10;$ $b) y = x^2, y = 2x^2, x = 1;$ $c) x = (y - 2)^2, x = 4y - 8$	22	$a) y = x^2 - 3x - 1, y = 2 - x;$ $b) y = 5 - x^2, x = 0, y = 0;$ $c) y = \sin 2x \cos^5 x,$ $y = 0, \left(0 \leq x \leq \frac{\pi}{2}\right)$
23	$a) y = -x^2 + 2x + 3, y = 7x - 3;$ $b) xy = 1, x = 1, x = 2, y = 0;$ $c) y = \frac{x}{(x^2 + 1)^2}, y = 0, x = 1$	24	$a) y = x^2 + 7x + 1, y = 5x + 4;$ $b) y = x^{-1}, x = 2, x = 4, y = 0;$ $c) x = 4 - y^2, x = y^2 - y$
25	$a) y = 4 - x^2, y = x + 2;$ $b) y = x, y = 5x^{-1}, x = 3;$ $c) x = \frac{1}{y\sqrt{1 + \ln y}},$ $x = 0, y = 1, y = e^3$	26	$a) y = -x^2 + 8x + 1, y = 8x;$ $b) y = \lg x, x = 10, y = 0;$ $c) y = \frac{e^{1/x}}{x^2}, y = 0, x = 1, x = 2$
27	$a) y = x^2 + 4x, y = 3x + 6;$ $b) xy = 6, x + y = 5;$ $c) y = x^2\sqrt{16 - x^2},$ $y = 0, (0 \leq x \leq 4)$	28	$a) y = 3x^2 - x, y = -2x^2 + 4x;$ $b) y = x^2, y = 9 - x^2;$ $c) x = \sqrt{4 - y^2}, x = 0, y = 0, y = 1$
29	$a) y = -x^2 - 7x, y = -11x + 3;$ $b) y = \sin 2x, x \in \left[0; \frac{\pi}{2}\right], y = 0;$ $c) y = (x - 1)^2, y^2 = x - 1$	30	$a) y = 2x^2, y = 3 - x^2;$ $b) y = 3^x, y = 3^{2x}, x = 1;$ $c) x = 4 - (y - 1)^2,$ $x = y^2 - 4x + 3$

Задание 1 Найти площади фигур, ограниченных линиями, заданными уравнениями.

Вар-т	Задание	Вар-т	Задание	Вар-т	Задание
1	$\begin{cases} x = 4\sqrt{2} \cos^3 t, \\ y = 2\sqrt{2} \sin^3 t, \\ x = 2(x \geq 2) \end{cases}$	2	$\begin{cases} x = \sqrt{2} \cos t, \\ y = 2\sqrt{2} \sin t, \\ y = 2(y \geq 2) \end{cases}$	3	$\begin{cases} x = 4(t - \sin t), \\ y = 4(1 - \cos t), \\ y = 4(0 < x < 8\pi, y \geq 4) \end{cases}$
4	$\begin{cases} x = 16 \cos^3 t, \\ y = 2 \sin^3 t, \\ x = 2(x \geq 2) \end{cases}$	5	$\begin{cases} x = 2 \cos t, \\ y = 6 \sin t, \\ y = 3(y \geq 3) \end{cases}$	6	$\begin{cases} x = 2(t - \sin t), \\ y = 2(1 - \cos t), \\ y = 3(0 < x < 4\pi, y \geq 3) \end{cases}$
7	$\begin{cases} x = 16 \cos^3 t, \\ y = \sin^3 t, \\ x = 6\sqrt{3}(x \geq 6\sqrt{3}) \end{cases}$	8	$\begin{cases} x = 6 \cos t, \\ y = 2 \sin t, \\ y = \sqrt{3}(y \geq \sqrt{3}) \end{cases}$	9	$\begin{cases} x = 3(t - \sin t), \\ y = 3(1 - \cos t), \\ y = 3(0 < x < 6\pi, y \geq 3) \end{cases}$
10	$\begin{cases} x = 8\sqrt{2} \cos^3 t, \\ y = \sqrt{2} \sin^3 t, \\ x = 4(x \geq 4) \end{cases}$	11	$\begin{cases} x = 2\sqrt{2} \cos t, \\ y = 3\sqrt{2} \sin t, \\ y = 3(y \geq 3) \end{cases}$	12	$\begin{cases} x = 6(t - \sin t), \\ y = 6(t - \cos t), \\ y = 9(0 < x < 12\pi, y \geq 9) \end{cases}$
13	$\begin{cases} x = 32 \cos^3 t, \\ y = \sin^3 t, \\ x = 4(x \geq 4) \end{cases}$	14	$\begin{cases} x = 3 \cos t, \\ y = 8 \sin t, \\ y = 4(y \geq 4) \end{cases}$	15	$\begin{cases} x = 6(t - \sin t), \\ y = 6(1 - \cos t), \\ y = 6(0 < x < 12\pi, y \geq 6) \end{cases}$
16	$\begin{cases} x = 8 \cos^3 t, \\ y = 4 \sin^3 t, \\ x = 3\sqrt{3}(x \geq 3\sqrt{3}) \end{cases}$	17	$\begin{cases} x = 6 \cos t, \\ y = 4 \sin t, \\ y = 2\sqrt{3}(y \geq 2\sqrt{3}) \end{cases}$	18	$\begin{cases} x = 10(t - \sin t), \\ y = 10(1 - \cos t), \\ y = 15(0 < x < 20\pi, y \geq 15) \end{cases}$
19	$\begin{cases} x = 2\sqrt{2} \cos^3 t, \\ y = \sqrt{2} \sin^3 t, \\ x = 1(x \geq 1) \end{cases}$	20	$\begin{cases} x = \sqrt{2} \cos t, \\ y = 4\sqrt{2} \sin t, \\ y = 4(y \geq 4) \end{cases}$	21	$\begin{cases} x = t - \sin t, \\ y = 1 - \cos t, \\ y = 1(0 < x < 2\pi, y \geq 1) \end{cases}$
22	$\begin{cases} x = 8 \cos^3 t, \\ y = 8 \sin^3 t, \\ x = 1(x \geq 1) \end{cases}$	23	$\begin{cases} x = 9 \cos t, \\ y = 4 \sin t, \\ y = 2(y \geq 2) \end{cases}$	24	$\begin{cases} x = 8(t - \sin t), \\ y = 8(1 - \cos t), \\ y = 12(0 < x < 16\pi, y \geq 12) \end{cases}$
25	$\begin{cases} x = 24 \cos^3 t, \\ y = 2 \sin^3 t, \\ x = 9\sqrt{3}(x \geq 9\sqrt{3}) \end{cases}$	26	$\begin{cases} x = 3 \cos t, \\ y = 8 \sin t, \\ x = 4\sqrt{3}(y \geq 4\sqrt{3}) \end{cases}$	27	$\begin{cases} x = 2(t - \sin t), \\ y = 2(1 - \cos t), \\ y = 2(0 < x < 4\pi, y \geq 2) \end{cases}$
28	$\begin{cases} x = 4\sqrt{2} \cos^3 t, \\ y = \sqrt{2} \sin^3 t, \\ x = 2(x \geq 2) \end{cases}$	29	$\begin{cases} x = 2\sqrt{2} \cos t, \\ y = 5\sqrt{2} \sin t, \\ y = 5(y \geq 5) \end{cases}$	30	$\begin{cases} x = 4(t - \sin t), \\ y = 4(1 - \cos t), \\ y = 6(0 < x < 8\pi, y \geq 6) \end{cases}$

Задание 3 Найти площади фигур, ограниченных линиями, заданными уравнениями в полярных координатах.

Вар-т	Задание	Вар-т	Задание
1	a) $r = 2 \sin 4\varphi$, b) $r = 4 \cos \varphi, r = 2 (r \geq 2)$	2	a) $r = \cos 2\varphi$, b) $r = 4 \sin \varphi, r = 2 (r \geq 2)$
3	a) $r = \frac{1}{2} + \sqrt{2} \cos \varphi$, b) $r = 4 \sin 3\varphi, r = 2 (r \geq 2)$	4	a) $r = \frac{1}{2} + \sqrt{2} \sin \varphi$, b) $r = \sqrt{3} \cos \varphi, r = \sin \varphi (0 \leq \varphi \leq \pi / 2)$
5	a) $r = 2 + \cos \varphi$, b) $r = 3 \cos \varphi, r = 1 (r \geq 1)$	6	a) $r = 2 + \sin \varphi$, b) $r = 2 \cos \varphi, r = 2\sqrt{3} \sin \varphi, (0 \leq \varphi \leq \pi / 2)$
7	a) $r = \sin 3\varphi$, b) $r = 6 \cos 3\varphi, r = 3 (r \geq 3)$	8	a) $r = \cos 3\varphi$, b) $r = 6 \sin 3\varphi, r = 3 (r \geq 3)$
9	a) $r = \sin 4\varphi$, b) $r = \cos \varphi, r = \sqrt{2} \cos(\varphi - \pi / 4)$ $(-\pi / 4 \leq \varphi \leq \pi / 2)$	10	a) $r = \cos 4\varphi$, b) $r = \sin \varphi, r = \sqrt{2} \cos(\varphi - \pi / 4)$ $(0 \leq \varphi \leq 3\pi / 4)$
11	a) $r = \frac{1}{2} \cos \varphi$, b) $r = 6 \cos 3\varphi, r = 3 (r \geq 3)$	12	a) $r = \frac{1}{2} \sin \varphi$, b) $r = 6 \sin 3\varphi, r = 3, (r \geq 3)$
13	a) $r^2 = 9 \sin 2\varphi$, b) $r = \cos \varphi, r = \sin \varphi (0 \leq \varphi \leq \pi / 2)$	14	a) $r^2 = 4 \cos 2\varphi$, b) $r = \sqrt{2} \cos(\varphi - \pi / 4)$, $r = \sqrt{2} \sin(\varphi - \pi / 4), (\pi / 4 \leq \varphi \leq 3\pi / 4)$
15	a) $r = 4 \sin 4\varphi$, b) $r = \sin \varphi, r = 2 \cos \varphi$	16	a) $r = 2 \cos 4\varphi$, b) $r = \cos \varphi, r = 2 \sin \varphi$
17	a) $r = \frac{1}{2} + \sin \varphi$, b) $r = 6 \cos \varphi, r = 4 \cos \varphi$	18	a) $r = \frac{1}{2} + \cos \varphi$, b) $r = 6 \sin \varphi, r = 4 \cos \varphi$
19	a) $r = 1 + \sqrt{2} \sin \varphi$, b) $r = 5 \cos \varphi, r = 2 (r \geq 2)$	20	a) $r = 1 + \sqrt{2} \cos \varphi$, b) $r = 3 \sin \varphi, r = 2 (r \geq 2)$
21	a) $r = 2 \sin 4\varphi$, b) $r = 3 / 2 \cos \varphi, r = 5 / 2 \sin \varphi$	22	a) $r = 4 \cos 4\varphi$, b) $r = 5 / 2 \sin \varphi, r = 3 / 2 \cos \varphi$
23	a) $r = \cos 6\varphi$, b) $r = 4 \sin \varphi, r = 2 (r \geq 2)$	24	a) $r = \sin 6\varphi$, b) $r = 2 \cos \varphi, r = 3 \sin \varphi$
25	a) $r = \cos \varphi + \sin \varphi$, b) $r = 2 + \cos \varphi$	26	a) $r = 2 \sin 4\varphi$, b) $r = 1 + \sin \varphi$
27	a) $r = 2 \cos 6\varphi$, b) $r = 3 \sin \varphi, r = 5 \cos \varphi$	28	a) $r = \cos \varphi - \sin \varphi$, b) $r = 3 \cos \varphi, r = 5 \sin \varphi$
29	a) $r = 1 + \cos \varphi$, b) $r = 3 \sin 3\varphi, r = 2 (r \geq 2)$	30	a) $r = 2(1 + \cos \varphi)$, b) $r = 2 \sin \varphi, r = 4 \sin \varphi$

Задание 4 Найти длины дуг кривых, заданных уравнениями в прямоугольной системе координат.

Вар-т	Задание	Вар-т	Задание
1	$y = \ln x, \sqrt{3} \leq x \leq \sqrt{15}$	2	$y = \frac{x^2}{4} - \frac{\ln x}{2}, 1 \leq x \leq 2$
3	$y = \sqrt{1-x^2} + \arcsin x, 0 \leq x \leq \frac{7}{9}$	4	$y = \ln \frac{5}{2x}, \sqrt{3} \leq x \leq \sqrt{8}$
5	$y = -\ln \cos x, 0 \leq x \leq \pi/6$	6	$y = e^x + 6, \ln \sqrt{8} \leq x \leq \ln \sqrt{15}$
7	$y = 2 + \arcsin \sqrt{x} + \sqrt{x-x^2}, \frac{1}{4} \leq x \leq 1$	8	$y = \ln(x^2 - 1), 2 \leq x \leq 3$
9	$y = \sqrt{1-x^2} + \arcsin x, 0 \leq x \leq \frac{8}{9}$	10	$y = \ln(1-x^2), 0 \leq x \leq \frac{1}{4}$
11	$y = 2 + \operatorname{ch} x, 0 \leq x \leq 1$	12	$y = 1 - \ln \cos x, 0 \leq x \leq \pi/6$
13	$y = e^x + 13, \ln \sqrt{15} \leq x \leq \ln \sqrt{24}$	14	$y = -\arccos \sqrt{x} + \sqrt{x-x^2}, 0 \leq x \leq \frac{1}{4}$
15	$y = 2 - e^x, \ln \sqrt{3} \leq x \leq \ln \sqrt{8}$	16	$y = \arcsin x - \sqrt{1-x^2}, 0 \leq x \leq \frac{15}{16}$
17	$y = 1 - \ln \sin x, \pi/3 \leq x \leq \pi/2$	18	$y = 1 - \ln(x^2 - 1), 3 \leq x \leq 4$
19	$y = \sqrt{x-x^2} - \arccos \sqrt{x} + 5, \frac{1}{9} \leq x \leq 1$	20	$y = -\arccos x + \sqrt{1-x^2} + 1, 0 \leq x \leq \frac{9}{16}$
21	$y = \ln \sin x, \pi/3 \leq x \leq \pi/2$	22	$y = \ln 7 - \ln x, \sqrt{3} \leq x \leq \sqrt{8}$
23	$y = \operatorname{ch} x + 3, 0 \leq x \leq 1$	24	$y = 1 + \arcsin x - \sqrt{1-x^2}, 0 \leq x \leq \frac{3}{4}$
25	$y = \ln \cos x + 2, 0 \leq x \leq \pi/6$	26	$y = e^x + 26, \ln \sqrt{8} \leq x \leq \ln \sqrt{24}$
27	$y = \frac{e^x + e^{-x}}{2} + 3, 0 \leq x \leq 2$	28	$y = \arccos \sqrt{x} - \sqrt{x-x^2} + 4, 0 \leq x \leq \frac{1}{2}$
29	$y = \frac{e^{2x} + e^{-2x} + 3}{3}, 0 \leq x \leq 2$	30	$y = e^x + e, \ln \sqrt{3} \leq x \leq \ln \sqrt{15}$

Задание 5 Найти длины дуг кривых, заданных параметрическими уравнениями.

Вар-т	Задание	Вар-т	Задание	Вар-т	Задание
1	$\begin{cases} x = 5(t - \sin t), \\ y = 5(1 - \cos t), \end{cases}$ $0 \leq t \leq \pi$	2	$\begin{cases} x = 3(2 \cos t - \cos 2t), \\ y = 3(2 \sin t - \sin 2t), \end{cases}$ $0 \leq t \leq 2\pi$	3	$\begin{cases} x = 4(\cos t + t \sin t), \\ y = 4(\sin t - t \cos t), \end{cases}$ $0 \leq t \leq 2$
4	$\begin{cases} x = (t^2 - 2) \sin t + 2t \cos t, \\ y = (2 - t^2) \cos t + 2t \sin t, \end{cases}$ $0 \leq t \leq \pi$	5	$\begin{cases} x = 10 \cos^3 t, \\ y = 10 \sin^3 t, \end{cases}$ $0 \leq t \leq \pi/2$	6	$\begin{cases} x = e^t (\cos t + \sin t), \\ y = e^t (\cos t - \sin t), \end{cases}$ $0 \leq t \leq \pi$
7	$\begin{cases} x = 3(t - \sin t), \\ y = 3(t - \cos t), \end{cases}$ $\pi \leq t \leq 2\pi$	8	$\begin{cases} x = \frac{1}{2} \cos t - \frac{1}{4} \cos 2t, \\ y = \frac{1}{2} \sin t - \frac{1}{4} \sin 2t, \end{cases}$ $\pi/2 \leq t \leq 2\pi/3$	9	$\begin{cases} x = 3(\cos t + t \sin t), \\ y = 3(\sin t - t \cos t), \end{cases}$ $0 \leq t \leq \pi/3$
10	$\begin{cases} x = (t^2 - 2) \sin t + 2t \cos t, \\ y = (2 - t^2) \cos t + 2t \sin t, \end{cases}$ $0 \leq t \leq \pi/3$	11	$\begin{cases} x = 6 \cos^3 t, \\ y = 6 \sin^3 t, \end{cases}$ $0 \leq t \leq \pi/3$	12	$\begin{cases} x = e^t (\cos t + \sin t), \\ y = e^t (\cos t - \sin t), \end{cases}$ $\pi/2 \leq t \leq \pi$
13	$\begin{cases} x = 2,5(t - \sin t), \\ y = 2,5(1 - \cos t), \end{cases}$ $\pi/2 \leq t \leq \pi$	14	$\begin{cases} x = 3,5(2 \cos t - \cos 2t), \\ y = 3,5(2 \sin t - \sin 2t), \end{cases}$ $0 \leq t \leq \pi/2$	15	$\begin{cases} x = 6(\cos t + t \sin t), \\ y = 6(\sin t - t \cos t), \end{cases}$ $0 \leq t \leq \pi$
16	$\begin{cases} x = (t^2 - 2) \sin t + 2t \cos t, \\ y = (2 - t^2) \cos t + 2t \sin t, \end{cases}$ $0 \leq t \leq \pi/2$	17	$\begin{cases} x = 8 \cos^3 t, \\ y = 8 \sin^3 t, \end{cases}$ $0 \leq t \leq \pi/6$	18	$\begin{cases} x = e^t (\cos t + \sin t), \\ y = e^t (\cos t - \sin t), \end{cases}$ $0 \leq t \leq 2\pi$
19	$\begin{cases} x = 4(t - \sin t), \\ y = 4(1 - \cos t), \end{cases}$ $\pi/2 \leq t \leq 2\pi/3$	20	$\begin{cases} x = 2(2 \cos t - \cos 2t), \\ y = 2(2 \sin t - \sin 2t), \end{cases}$ $0 \leq t \leq \pi/3$	21	$\begin{cases} x = 8(\cos t + t \sin t), \\ y = 8(\sin t - t \cos t), \end{cases}$ $0 \leq t \leq \pi/4$
22	$\begin{cases} x = (t^2 - 2) \sin t + 2t \cos t, \\ y = (2 - t^2) \cos t + 2t \sin t, \end{cases}$ $0 \leq t \leq 2\pi$	23	$\begin{cases} x = 4 \cos^3 t, \\ y = 4 \sin^3 t, \end{cases}$ $\pi/6 \leq t \leq \pi/4$	24	$\begin{cases} x = e^t (\cos t + \sin t), \\ y = e^t (\cos t - \sin t), \end{cases}$ $0 \leq t \leq 3\pi/2$
25	$\begin{cases} x = 2(t - \sin t), \\ y = 2(1 - \cos t), \end{cases}$ $0 \leq t \leq \pi/2$	26	$\begin{cases} x = 4(2 \cos t - \cos 2t), \\ y = 4(2 \sin t - \sin 2t), \end{cases}$ $0 \leq t \leq \pi$	27	$\begin{cases} x = 2(\cos t + t \sin t), \\ y = 2(\sin t - t \cos t), \end{cases}$ $0 \leq t \leq \pi/2$
28	$\begin{cases} x = (t^2 - 2) \sin t + 2t \cos t, \\ y = (2 - t^2) \cos t + 2t \sin t, \end{cases}$ $0 \leq t \leq 3\pi$	29	$\begin{cases} x = 2 \cos^3 t, \\ y = 2 \sin^3 t, \end{cases}$ $0 \leq t \leq \pi/4$	30	$\begin{cases} x = e^t (\cos t + \sin t), \\ y = e^t (\cos t - \sin t), \end{cases}$ $\pi/6 \leq t \leq \pi/4$

Задание 6 Найти длины дуг кривых, заданных уравнениями в полярных координатах.

Вар-т	Задание	Вар-т	Задание
1	$r = 3e^{3\varphi/4}, -\pi/2 \leq \varphi \leq \pi/2$	2	$r = 2e^{4\varphi/3}, -\pi/2 \leq \varphi \leq \pi/2$
3	$r = \sqrt{2}e^\varphi, -\pi/2 \leq \varphi \leq \pi/2$	4	$r = 5e^{5\varphi/12}, -\pi/2 \leq \varphi \leq \pi/2$
5	$r = 6e^{12\varphi/5}, -\pi/2 \leq \varphi \leq \pi/2$	6	$r = 3e^{3\varphi/4}, 0 \leq \varphi \leq \pi/3$
7	$r = 4e^{4\varphi/3}, 0 \leq \varphi \leq \pi/3$	8	$r = \sqrt{2}e^\varphi, 0 \leq \varphi \leq \pi/3$
9	$r = 5e^{5\varphi/12}, 0 \leq \varphi \leq \pi/3$	10	$r = 12e^{12\varphi/5}, 0 \leq \varphi \leq \pi/3$
11	$r = 1 - \sin \varphi, -\pi/2 \leq \varphi \leq -\pi/6$	12	$r = 2(1 - \cos \varphi), -\pi \leq \varphi \leq -\pi/2$
13	$r = 3(1 + \sin \varphi), -\pi/6 \leq \varphi \leq 0$	14	$r = 4(1 - \sin \varphi), 0 \leq \varphi \leq \pi/6$
15	$r = 5(1 - \cos \varphi), -\pi/3 \leq \varphi \leq 0$	16	$r = 6(1 + \sin \varphi), -\pi/2 \leq \varphi \leq 0$
17	$r = 7(1 - \sin \varphi), -\pi/6 \leq \varphi \leq \pi/6$	18	$r = 8(1 - \cos \varphi), -2\pi/3 \leq \varphi \leq 0$
19	$r = 2\varphi, 0 \leq \varphi \leq 3/4$	20	$r = 2\varphi, 0 \leq \varphi \leq 4/3$
21	$r = 2\varphi, 0 \leq \varphi \leq 5/12$	22	$r = 2\varphi, 0 \leq \varphi \leq 12/5$
23	$r = 4\varphi, 0 \leq \varphi \leq 3/4$	24	$r = 3\varphi, 0 \leq \varphi \leq 4/3$
25	$r = 5\varphi, 0 \leq \varphi \leq 12/5$	26	$r = 2 \cos \varphi, 0 \leq \varphi \leq \pi/6$
27	$r = 8 \cos \varphi, 0 \leq \varphi \leq \pi/4$	28	$r = 6 \cos \varphi, 0 \leq \varphi \leq \pi/3$
29	$r = 2 \sin \varphi, 0 \leq \varphi \leq \pi/6$	30	$r = 8 \sin \varphi, 0 \leq \varphi \leq \pi/4$

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Задание 7 Найти объем тела, полученного вращением указанных линий: в варианте а) вокруг OX , в варианте б) – вокруг OY . Выполнить чертеж.

Вар-т	Задание	Вар-т	Задание
1	а) $y = e^x, \quad x = 0, \quad x = 1, \quad y = 0;$ б) $y^2 - x^2 = -4, \quad y = \pm 2$	2	а) $y = x^2, \quad x = 2, \quad y = 0;$ б) $\begin{cases} x = 3(t - \sin t), \\ y = 3(1 - \cos t), \end{cases} \quad y \geq 3$
3	а) $y^2 = x, \quad x = 1;$ б) $\begin{cases} x = 2(t - \sin t), \\ y = 2(1 - \cos t), \end{cases} \quad 0 \leq t \leq \pi, \quad y = 0$	4	а) $y = x^2, \quad y = x;$ б) $\frac{x^2}{16} + \frac{y^2}{9} = 1, \quad y \geq 1$
5	а) $x = y^2, \quad y = x;$ б) $\frac{x^2}{4} + \frac{y^2}{9} = 1, \quad y \geq 2$	6	а) $\begin{cases} x = 4 \cos t, \\ y = 2 \sin t, \end{cases}$ б) $\frac{x^2}{9} + \frac{y^2}{16} = 1, \quad y \geq 2$
7	а) $y = x^2, \quad y = 2x;$ б) $\frac{x^2}{9} - \frac{y^2}{4} = 1, \quad y = \pm 2$	8	а) $y = e^x, \quad y = e^{2x}, \quad x = 1;$ б) $\begin{cases} x = 4 \cos t, \\ y = 2 \sin t, \end{cases} \quad y \geq 1$
9	а) $y = 2^x, \quad y = 2^{3x}, \quad x = 1;$ б) $\frac{x^2}{4} - \frac{y^2}{9} = 1, \quad y = \pm 2$	10	а) $y = \frac{1}{x}, \quad x = 1, \quad x = 3, \quad y = 0;$ б) $y = (x - 2)^3, \quad x = 4y - 8$
11	а) $xy = 5, \quad x = 1, \quad x = 2, \quad y = 0;$ б) $\begin{cases} x = 9 \cos t, \\ y = 4 \sin t, \end{cases} \quad y \geq 2$	12	а) $y = e^{2x}, \quad x = 0, \quad x = 1, \quad y =$ б) $y = x^2, \quad y = 0, \quad x = 2$
13	а) $y = 3^x, \quad y = 3^{2x}, \quad x = 1;$ б) $\frac{x^2}{4} - \frac{y^2}{16} = 1, \quad y = \pm 1$	14	а) $y = -x^2 + 1, \quad y = 0;$ б) $\begin{cases} x = 3(t - \sin t), \\ y = 3(1 - \cos t), \end{cases} \quad y = 0, 0 \leq x \leq \pi$
15	а) $y = x, \quad y = 2x, \quad x = 2;$ б) $\frac{x^2}{4} - \frac{y^2}{9} = 1, \quad y = \pm 3$	16	а) $y = x^2, \quad y = 2x^2, \quad x = 1;$ б) $\begin{cases} x = 3 \cos t, \\ y = \sqrt{2} \sin t, \end{cases} \quad y \geq 1$

17	a) $y = \cos x, y = 0, x = 0;$ b) $x = \sqrt{4 - y^2}, x = 0, y = 0, y = 1$	18	a) $y = \sin x, x = \pi/2, y = 0;$ b) $\begin{cases} x = 3 \cos t, \\ y = \sqrt{2} \sin t \end{cases}$
19	a) $y = \operatorname{tg} x, x = \pi/2, y = 0;$ b) $\begin{cases} x = 3 \cos t, \\ y = 2 \sin t, \end{cases} y \geq 1$	20	a) $y = x, y = 4x, x = 2;$ b) $xy = 6, x = 0, y = 6, y = 1$
21	a) $y = 2 \cos x, y = 0, x = 0;$ b) $x = 2\sqrt{y}, y = 4, x = 0$	22	a) $y = 3 \cos x, y = 0, x = 0;$ b) $\begin{cases} x = 2 \cos t, \\ y = 3 \sin t \end{cases}$
23	a) $y = -x^2 + 1, y = 0;$ b) $\begin{cases} x = \cos t, \\ y = 2 \sin t \end{cases}$	24	a) $y = 2 \sin x, x = \pi/2, y = 0;$ b) $\begin{cases} x = 2 \cos t, \\ y = 6 \sin t \end{cases}$
25	a) $y = \sin x, x = \pi/4, y = 0;$ b) $\begin{cases} x = 2 \cos t, \\ y = 6 \sin t, \end{cases} y \geq 3$	26	a) $y = x^2, y = 2x^2, x = 1;$ b) $\frac{x^2}{9} - \frac{y^2}{4} = 1, y = \pm 1$
27	a) $y = 2^x, y = 2^{2x}, x = 1;$ b) $\frac{x^2}{16} - \frac{y^2}{9} = 1, y = \pm 2$	28	a) $y = \operatorname{tg} 2x, x = \pi, y = 0;$ b) $x = 3\sqrt{y}, y = 9, x = 0$
29	a) $y = -x^2 + 2, y = 0;$ b) $\frac{x^2}{9} - \frac{y^2}{16} = 1, y = \pm 2$	30	a) $y = 2x^2, x = 1, y = 0;$ b) $\begin{cases} x = 2(t - \sin t), \\ y = 2(1 - \cos t), \end{cases} y \geq 2$